



Toward a More Robust Facial Emotion Recognition

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18794

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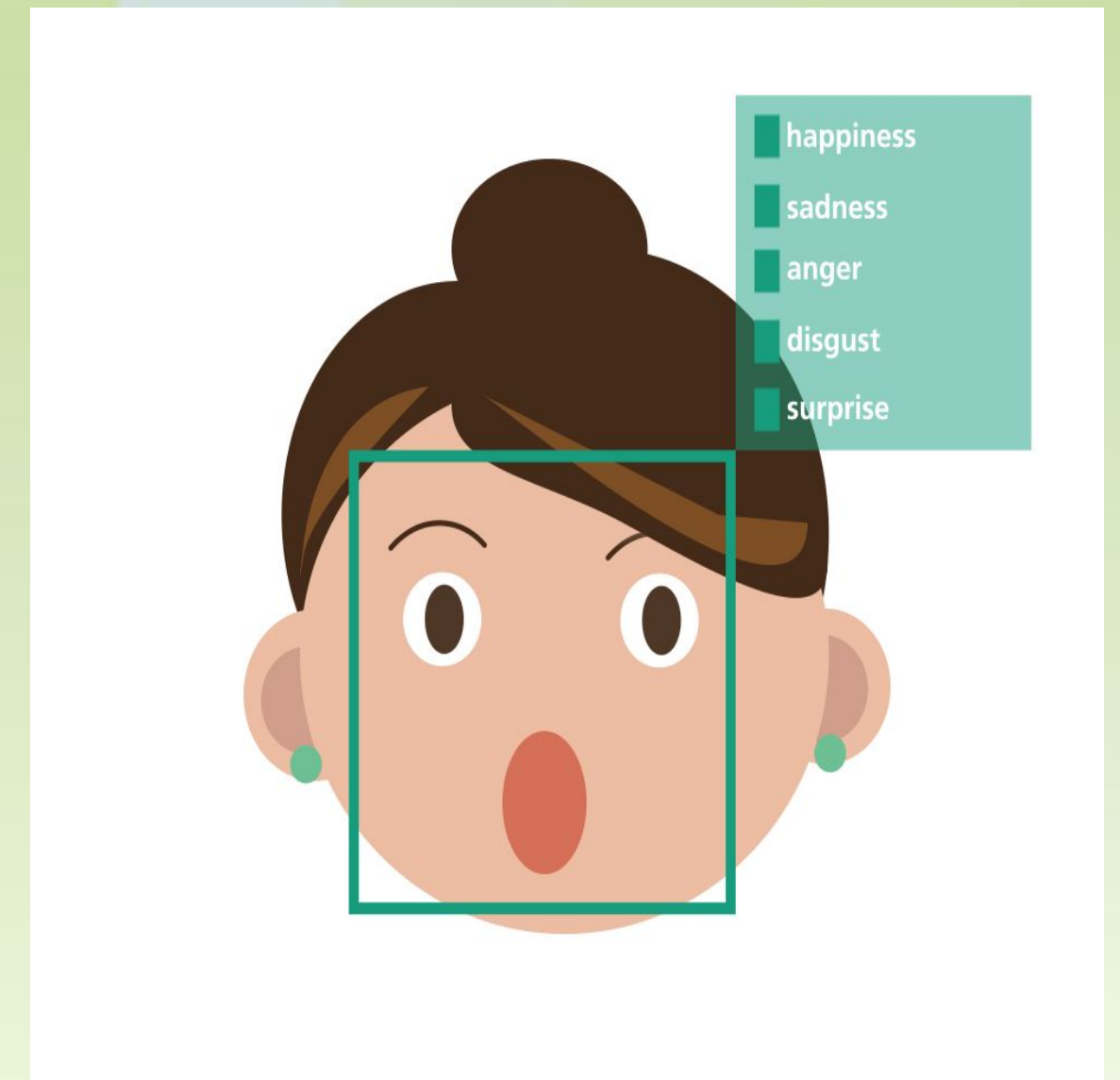
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Objectives

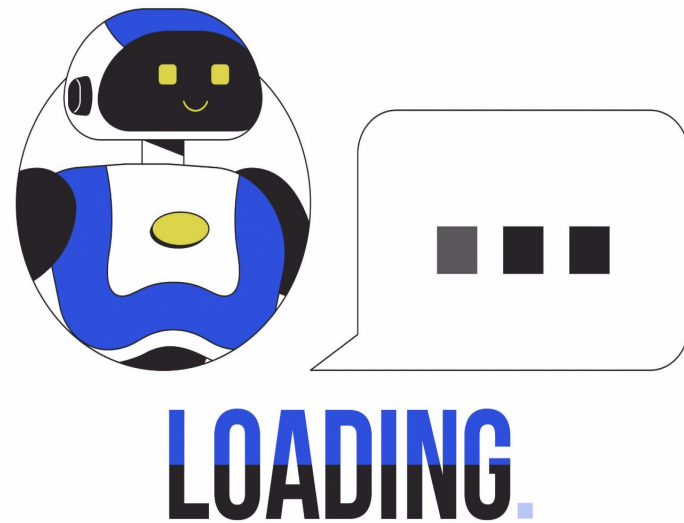
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Introduction

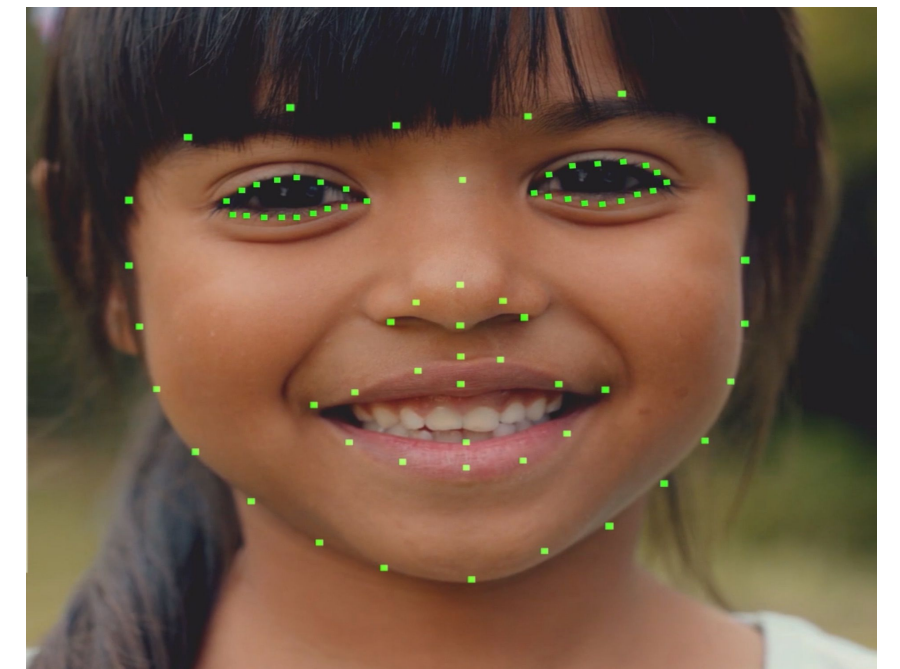
Motivation & Problem



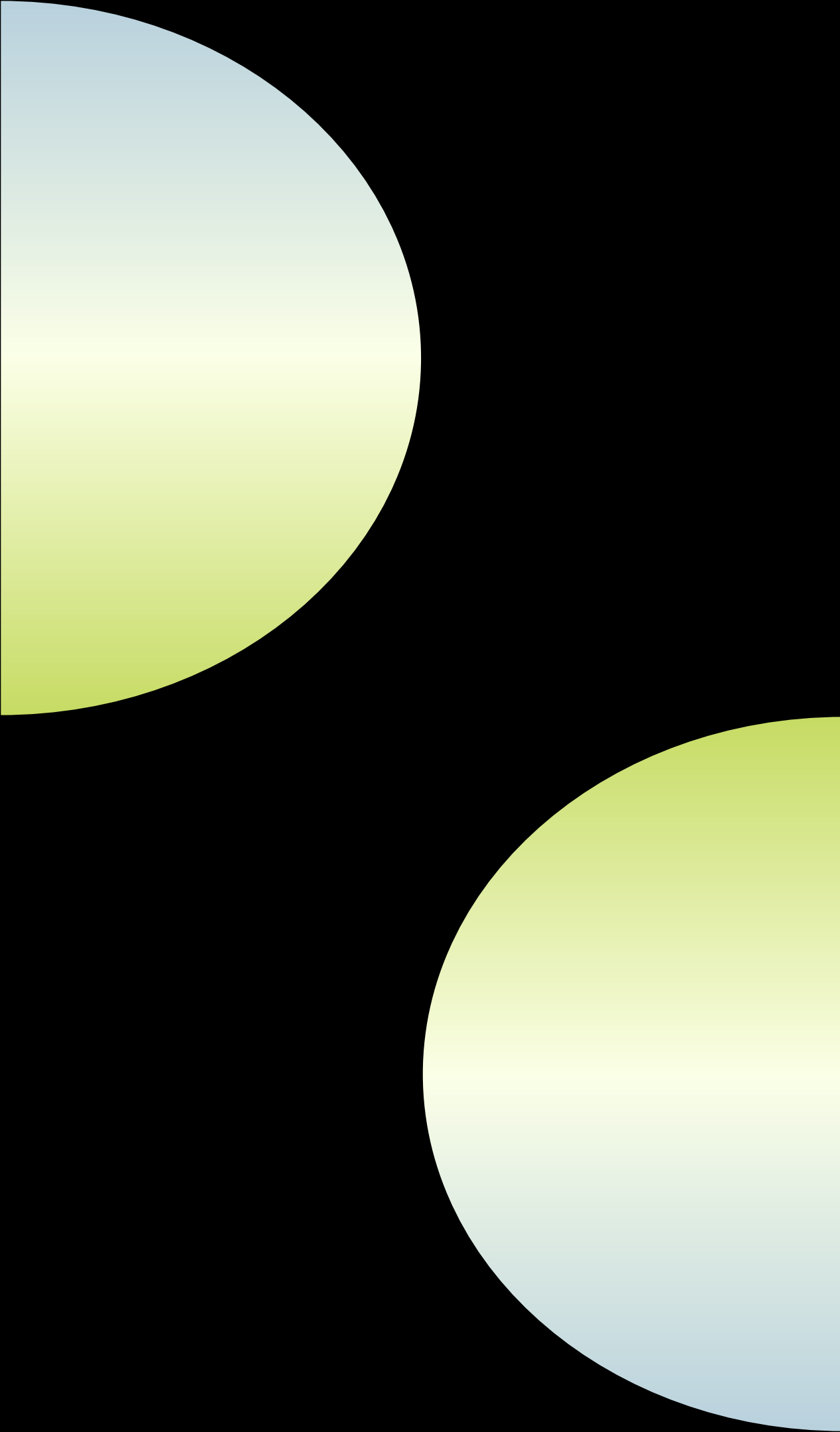
Traditional emotion recognition methods have rooms to improve on speed and accuracy



Current needs for emotion recognition usually locate in video/streaming conditions and require real-time performance

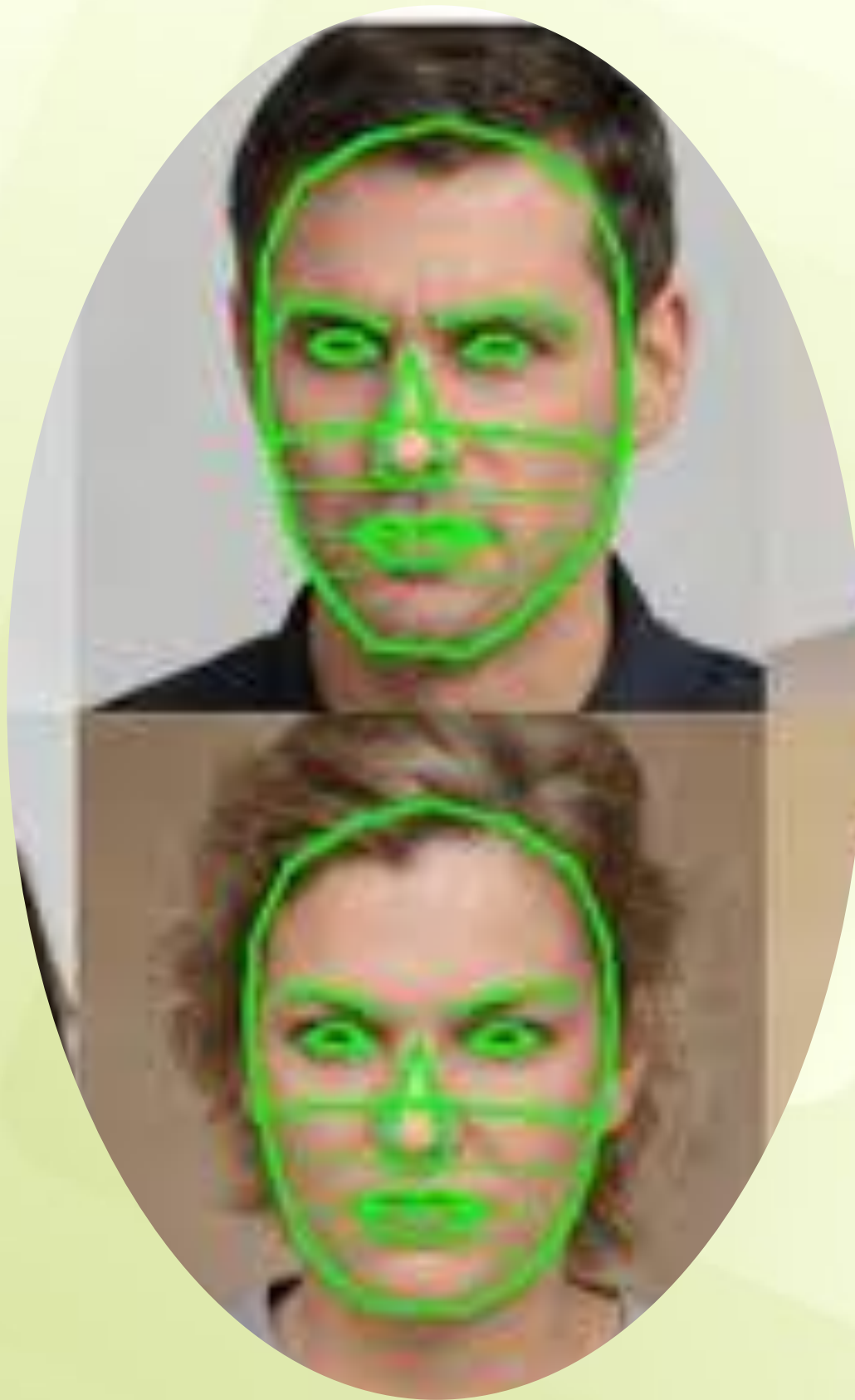


We take advantage of video/streaming conditions by scaling single-image prediction to multi-image, continuous prediction for better accuracy and use landmark feature extraction to reduce prediction workload



Our Approach

- Leverage accuracy of real-time facial emotion recognition with deep learning techniques
- Utilize landmark feature extraction to reduce input size for faster recognition
- Enhance dataset by Generative AI
- Sample frames from short video clips for multi-image continuous prediction to enhance accuracy



Why is FER important?

- Adaptive tutoring systems that respond to student frustration or confusion
- Driver safety monitoring detecting drowsiness or distraction
- Healthcare applications for pain assessment and mental health screening
- Human-robot interaction requiring emotional awareness

Objectives

Why is FER Hard?

Partial occlusions

Pose & lighting variations

Cross-subject
generalization

Subtle expressions
(near-neutral)

The Reality Gap: Why Current Systems Fail

Challenge

Impact on Recognition

Partial Occlusion

Glasses, hands, hair block key facial regions

Pose Variation

Side profiles lose critical landmark visibility

Lighting Changes

Shadows create false contours, wash out features

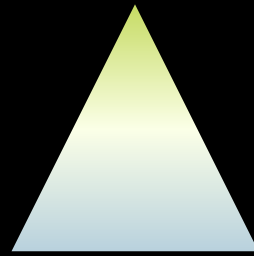
Subject Variability

My neutral face $\neq$ your neutral face

Subtle Expressions

Micro-expressions and low-amplitude emotions missed

Limitations of Existing Approaches

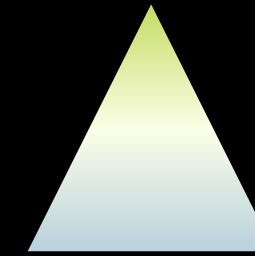


CNN-Based Models (Current State-of-the-Art)

- Learn features directly from pixel intensities
- Achieve high accuracy on benchmark datasets (80-90%)

Problems:

- unstable frame-by-frame predictions in video - flickering between emotions
- poor cross-domain generalization - train on dataset A, fail on dataset B
- sensitive to photometric variations (lighting, contrast, color balance)
- computationally expensive for real-time deployment



Sequence Models (RNN/LSTM on raw frames)

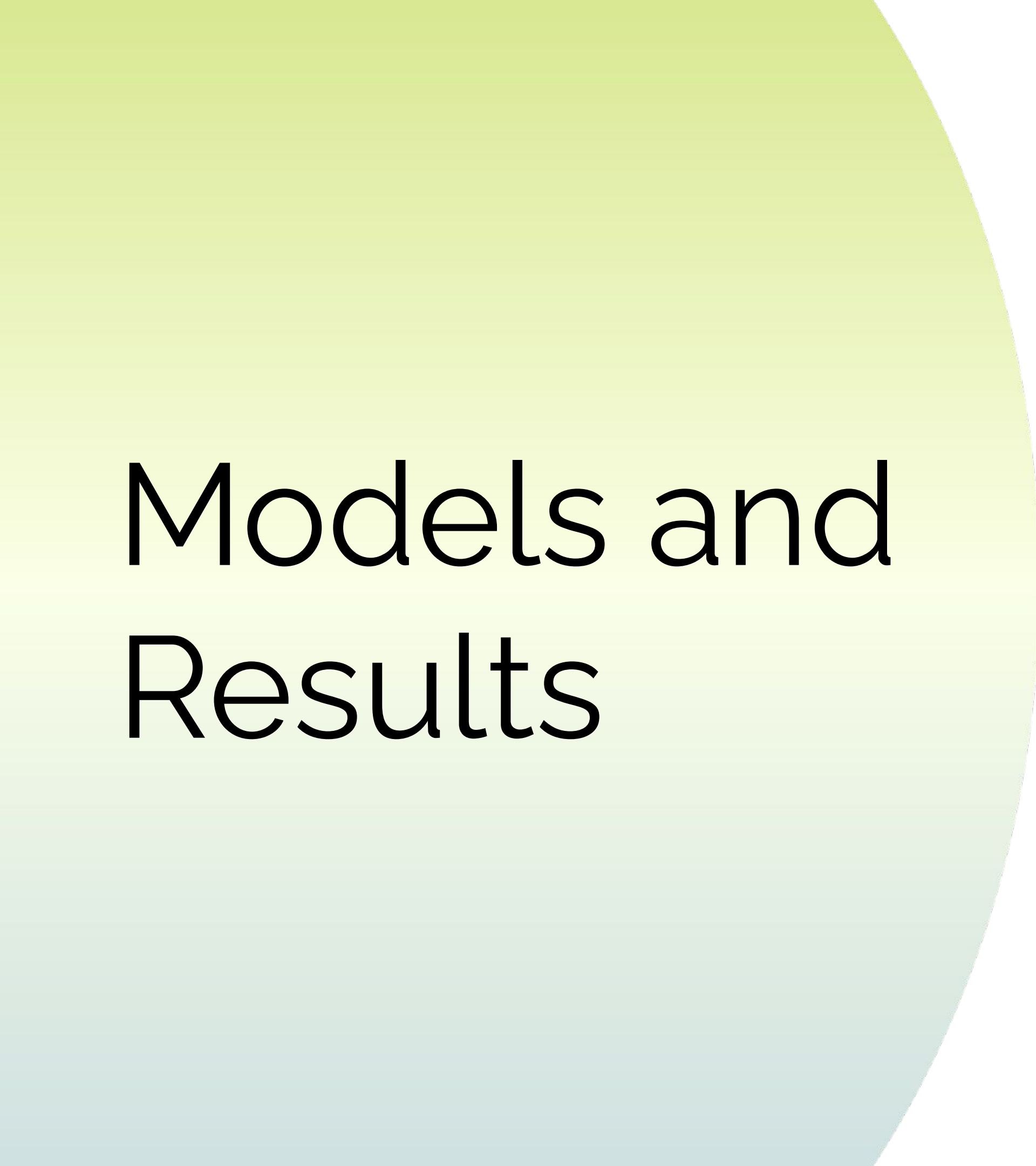
- Process video sequences to capture temporal dynamics
- Can improve prediction stability over time

Problems:

- operating on raw pixels requires massive computation
- still inherit CNN sensitivity to appearance variations

The Two Gaps We Address

1. Models don't account for individual facial morphology - the structural differences between faces that affect how emotions manifest
2. Single-image models miss the motion cues that distinguish subtle expressions - the difference between neutral and a building smile is visible in motion, not in a single frame



Models and Results

Baseline Model



Baseline CNN Model from Paper

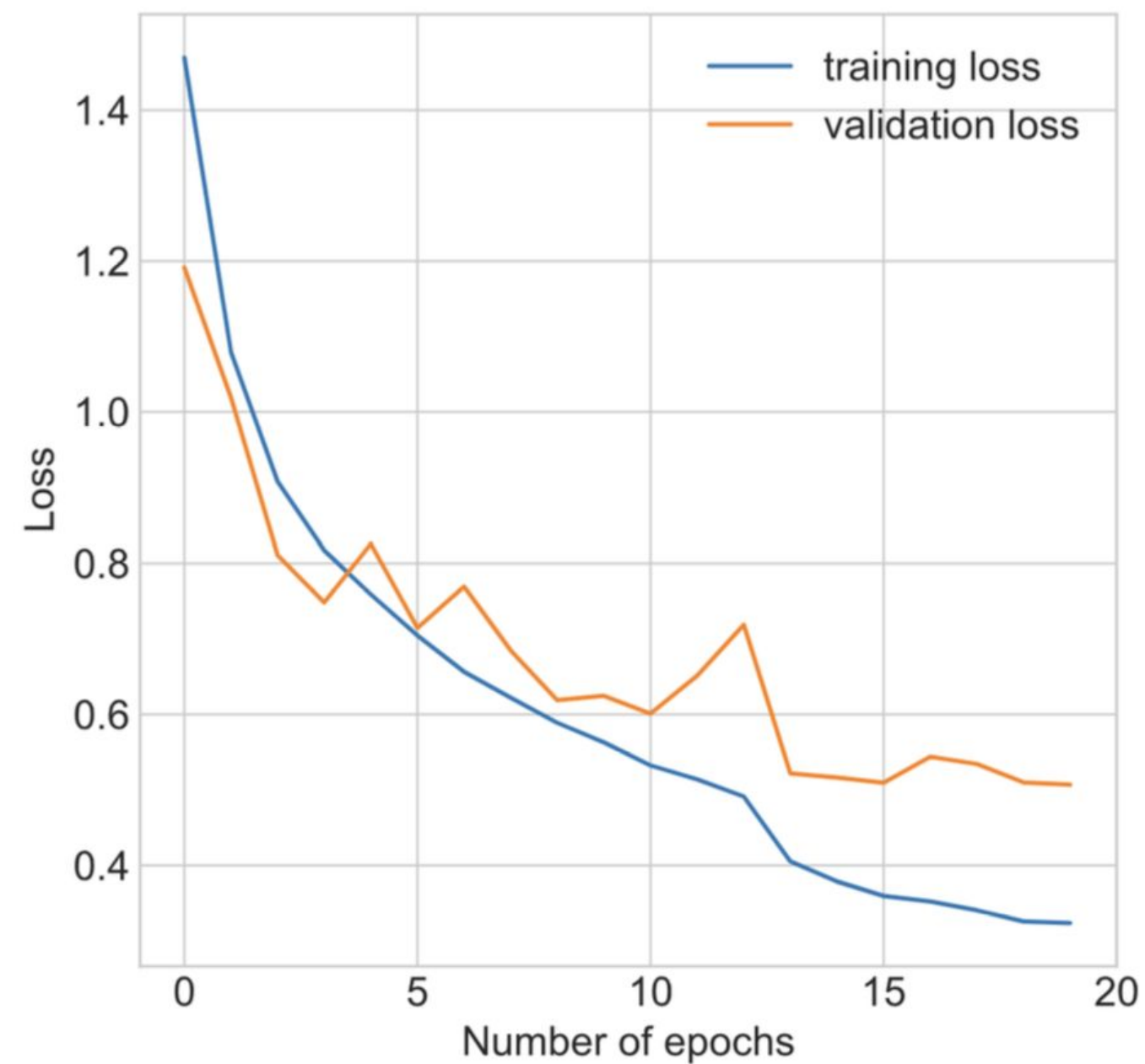


Figure 1: Train and validation loss of the baseline model for RAF-DB



(a) Prediction: Angry (b) Prediction: Disgust (c) Prediction: Happy

Figure 4: Sample Predictions from Baseline Model

Test Accuracy:
82.46%

Initial Experiment



Dataset for Single-Frame FER

RAF-DB: large-scale, "real-world" database containing 62,916 facial images of 75x75 pixels sourced from the internet



i. surprise

ii. sad

iii. neutral

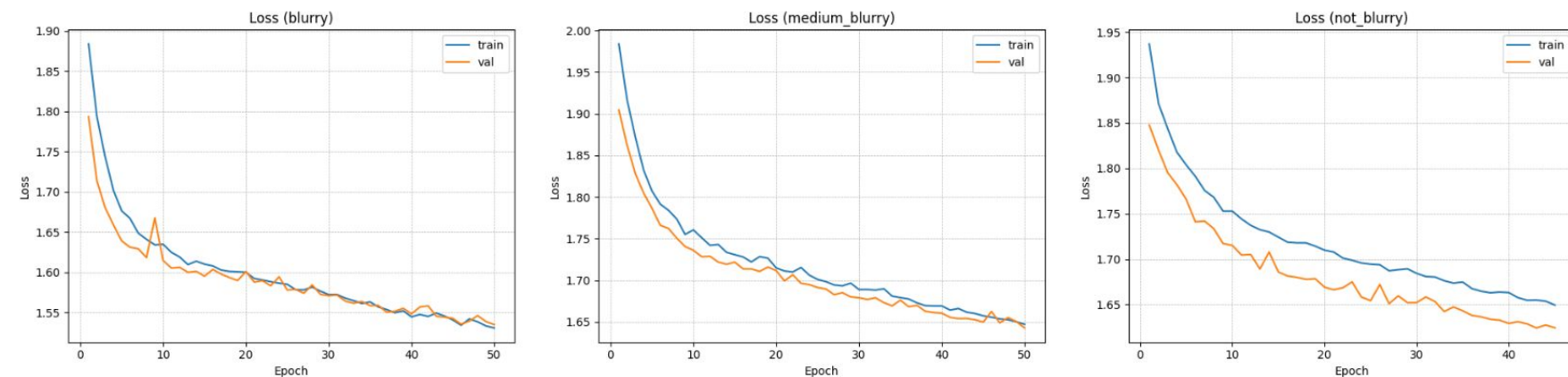
iv. happy

v. fear

vi. disgust

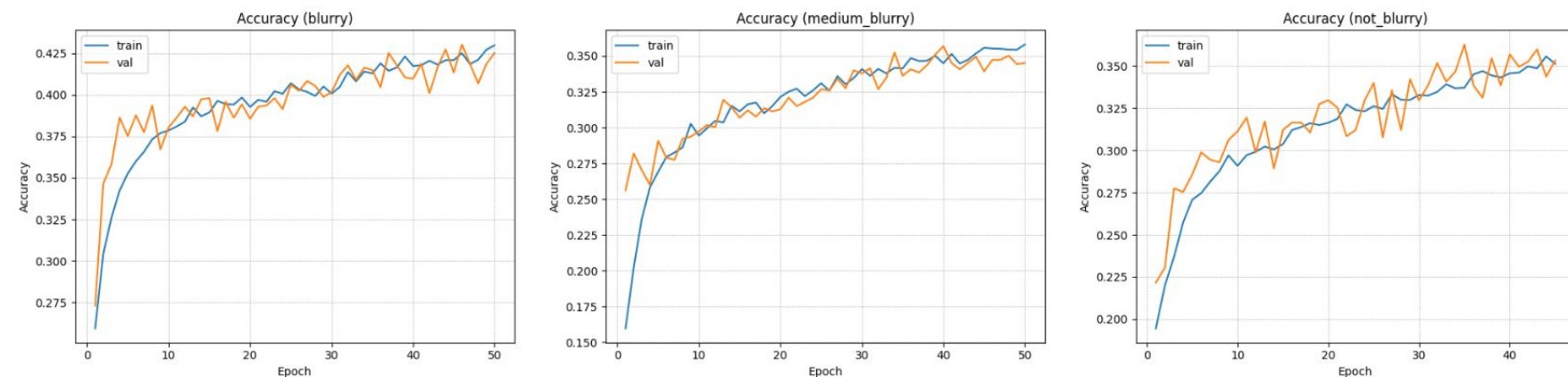
vii. angry

5 Landmarks + Blurity + MLP Model



(a) Train-Val Loss (Blurry) (b) Train-Val Loss (Mid Blurry) (c) Train-Val Loss (Not Blurry)

Figure 2: Train-Val Loss for the Models



(a) Accuracy (Blurry) (b) Accuracy (Mid Blurry) (c) Accuracy (Not Blurry)

Figure 3: Train-Val Accuracy for the Models

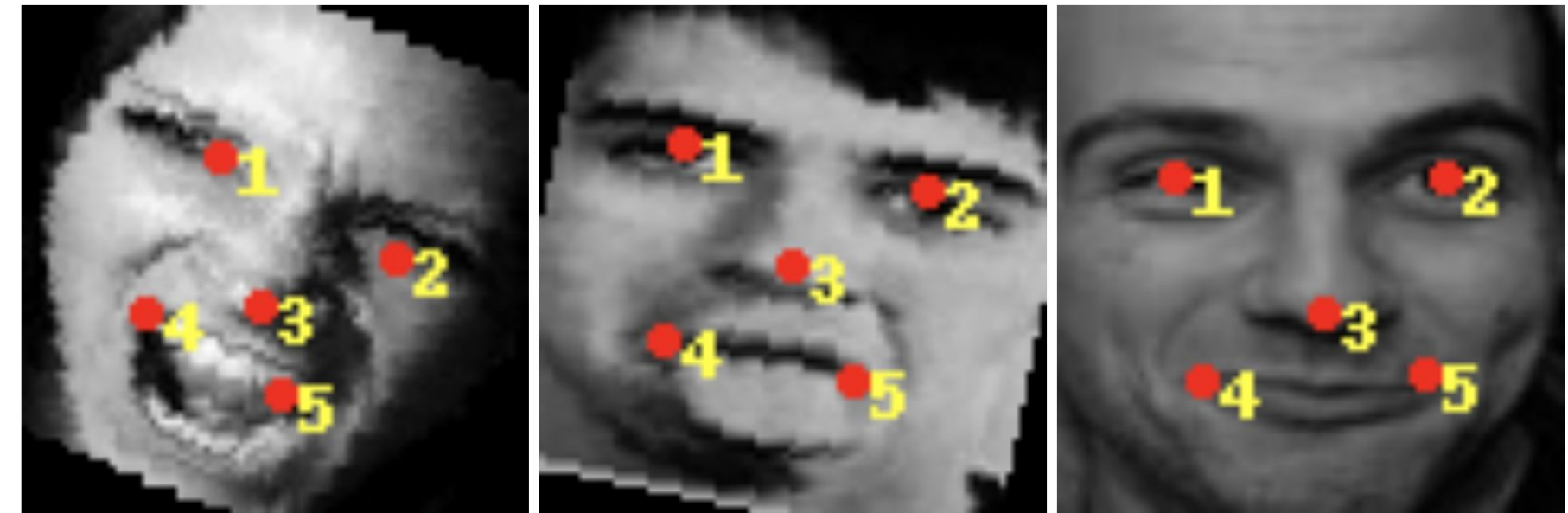


Figure 5: Sample Predictions from Our Models

Test Accuracy:
37.20%

Next-Stage Solution



Dataset for Single-Frame FER

RAF-DB: large-scale, "real-world" database containing 62,916 facial images of 75x75 pixels sourced from the internet



i. surprise

ii. sad

iii. neutral

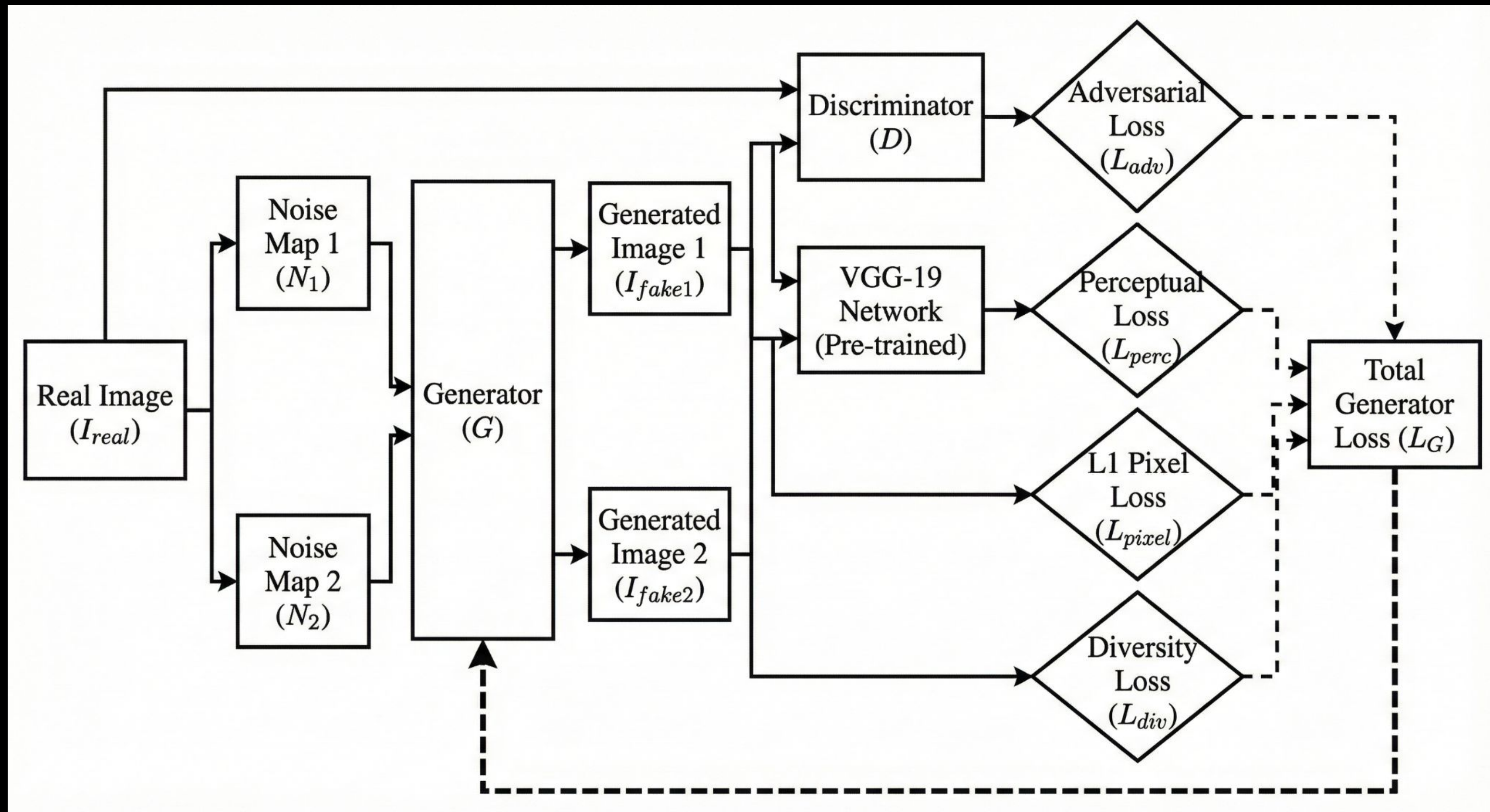
iv. happy

v. fear

vi. disgust

vii. angry

1. Train FaceID-GAN for Data Augmentation



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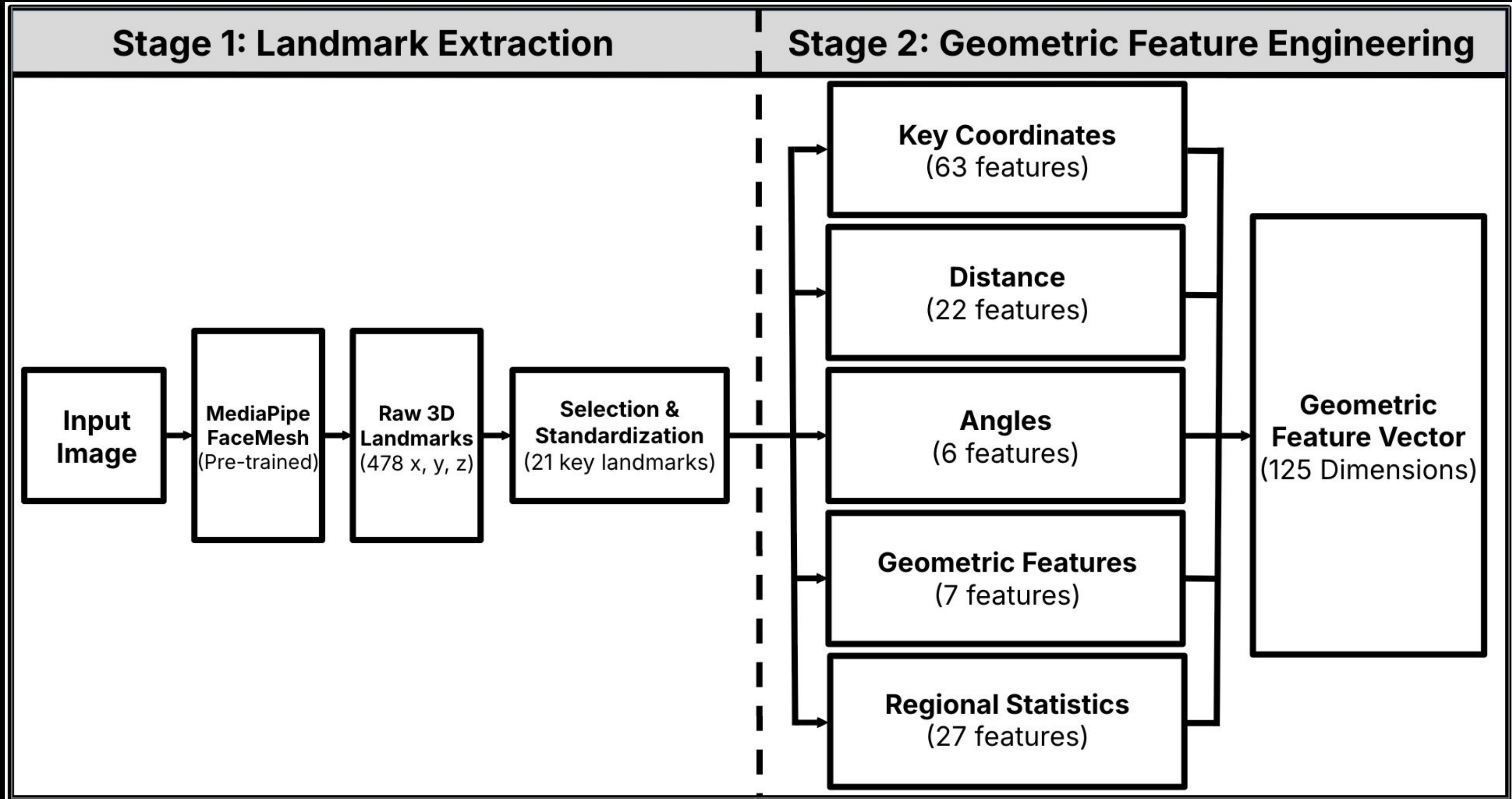








2. Facial Geometric Features Extraction



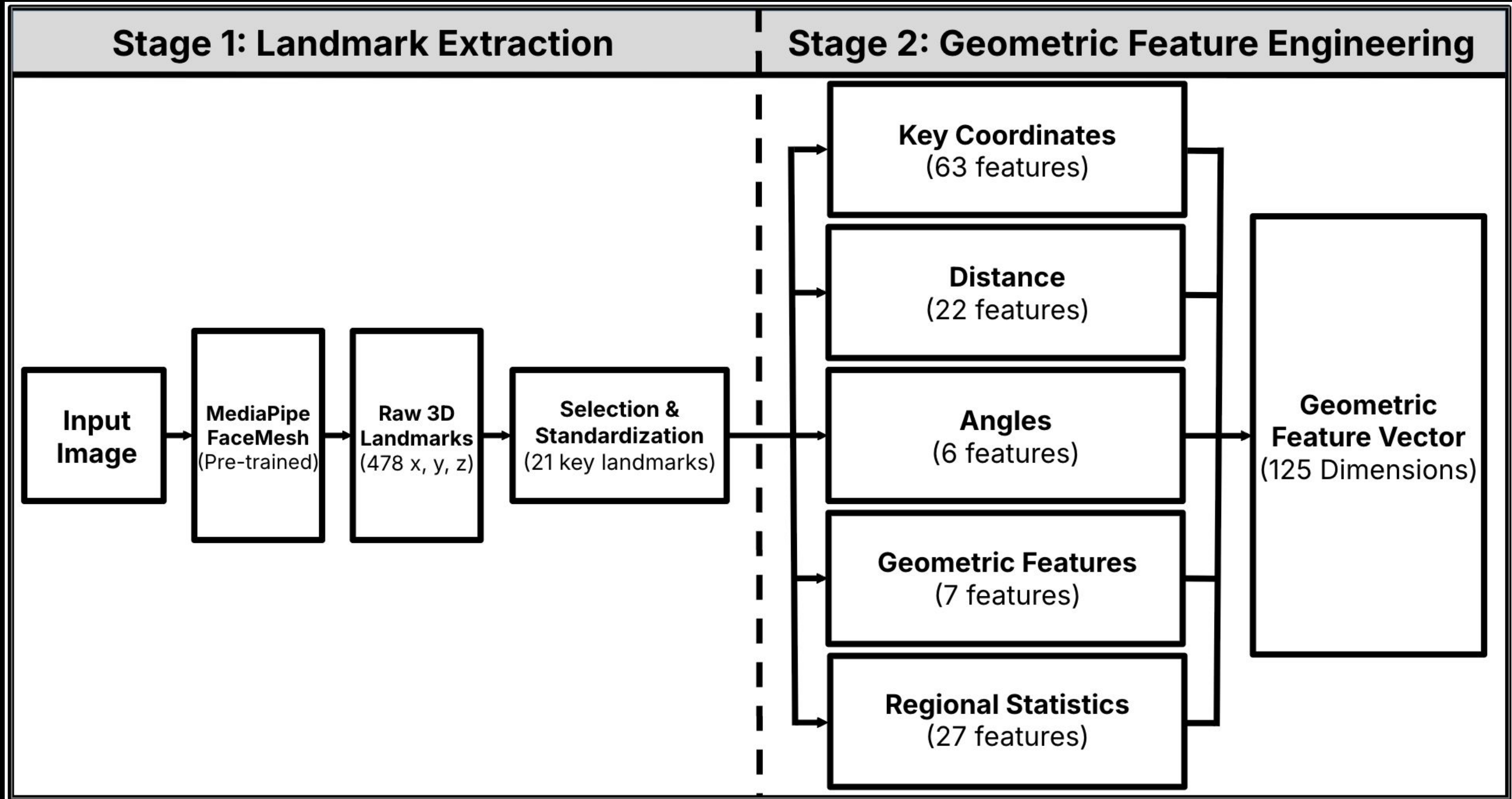
2. Facial Geometric Features Extraction

Stage 1: Landmarks Extraction

Deploy MediaPipe's FaceMesh and extract 478 3D facial landmarks (x, y, z) from each image. We pick the most critical 21 landmarks.



2. Facial Geometric Features Extraction



2. Facial Geometric Features Extraction (Cont'd)

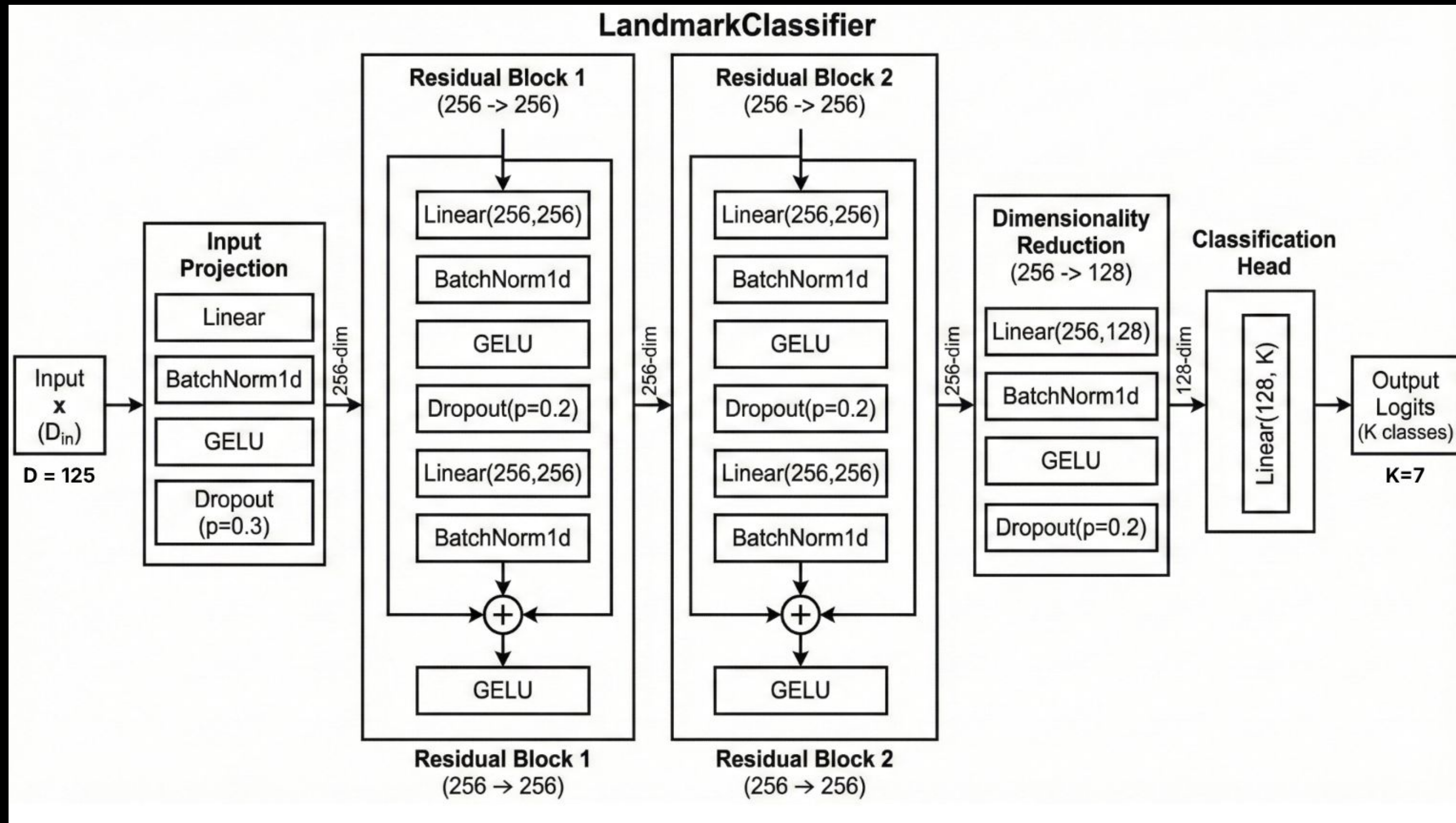
Stage 2: Geometric Feature Engineering

For the 21 landmarks, we perform various normalizations and scalings for pose invariance.

Then, we conduct 125 influential geometric features in 5 categories:

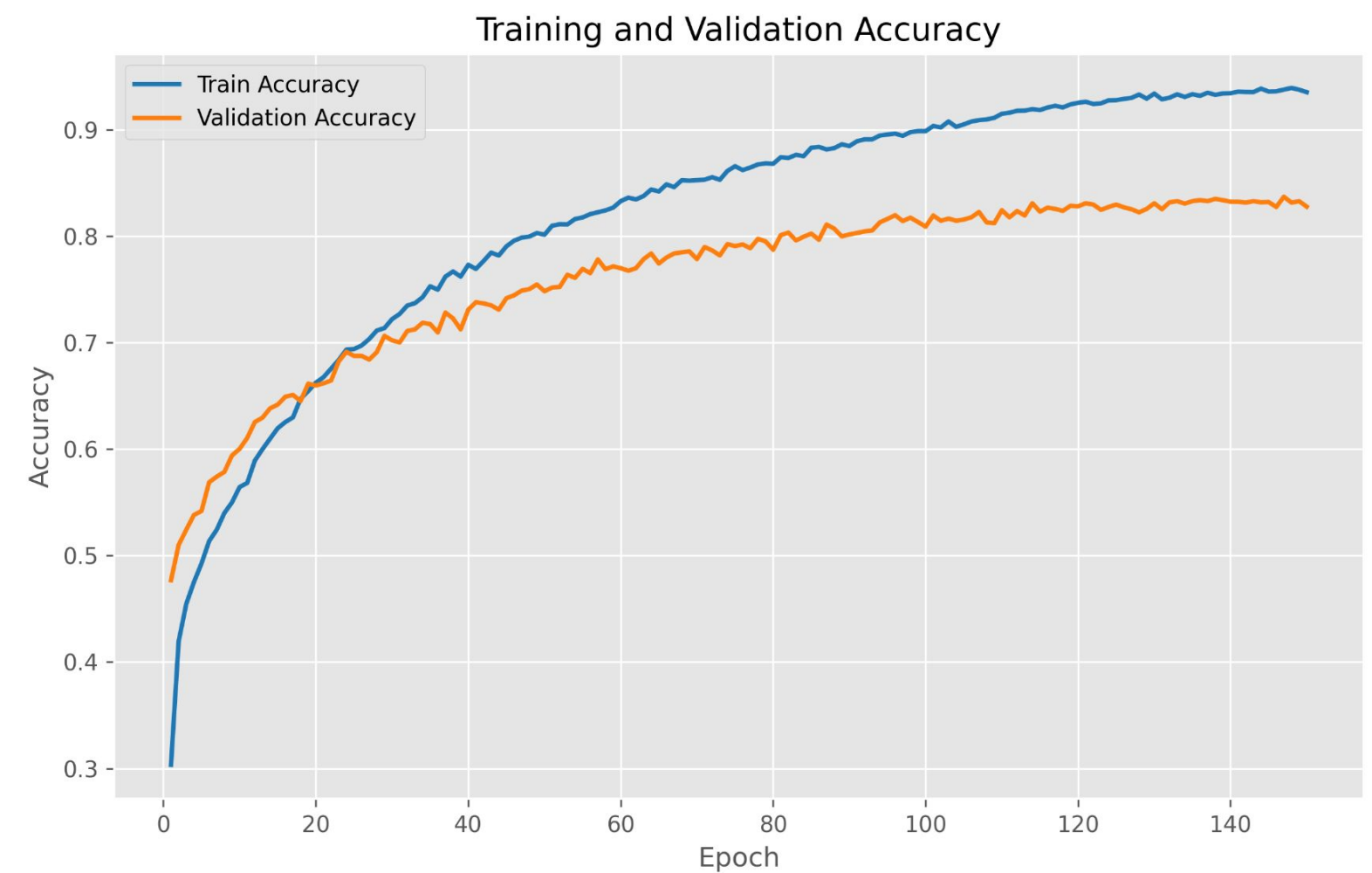
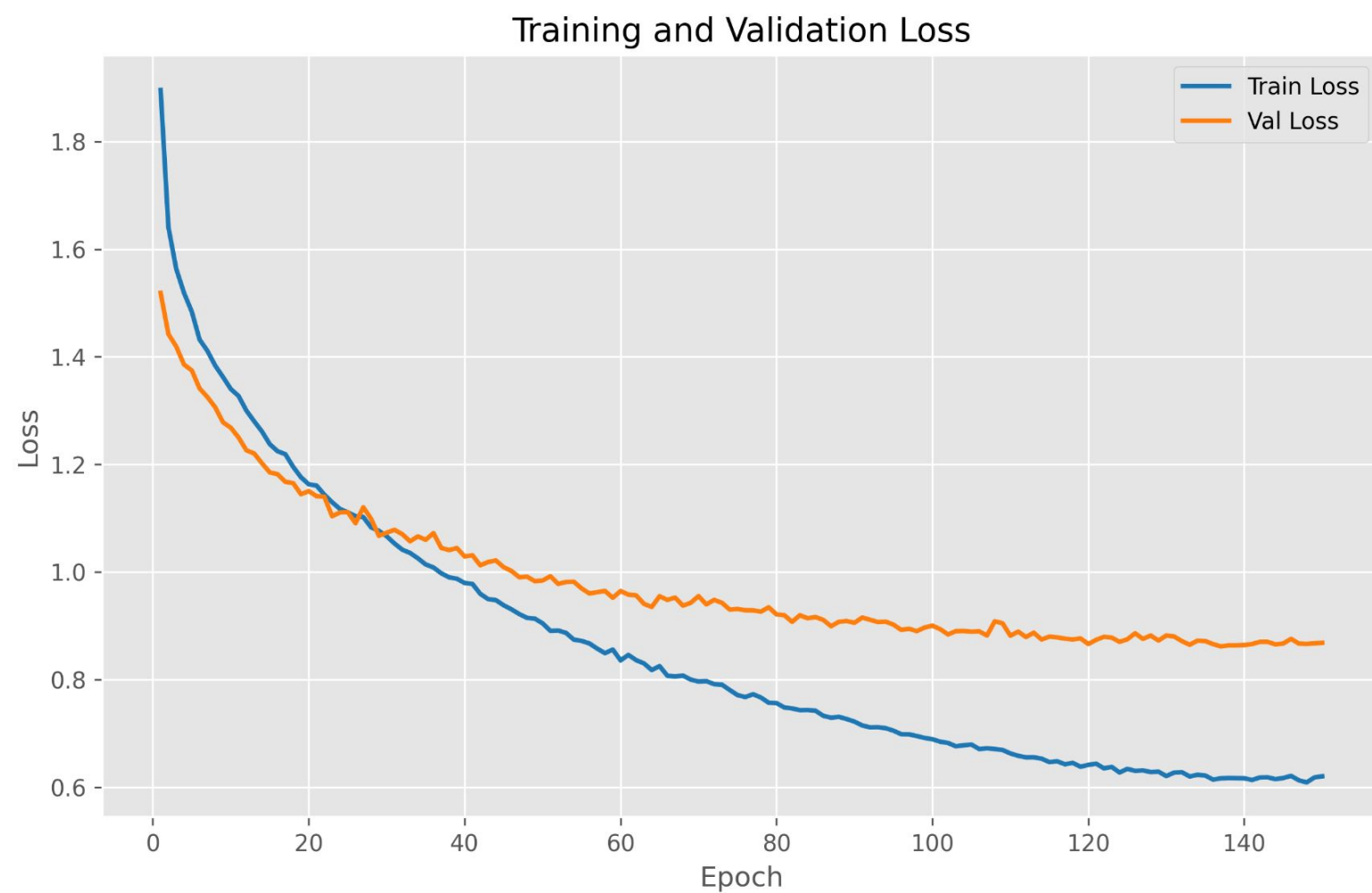
- **63 key coordinates** - 21 critical facial points (eyes, nose, mouth, eyebrows, chin)
- **22 distances** - facial dimensions between landmark pairs (eye width/height)
- **6 angles** - facial geometry from landmark triplets (eye-bridge angle, mouth angle)
- **7 geometric ratios** - proportional relationships (eye openness)
- **27 regional statistics** - 9 facial regions with spread, aspect ratio, and centroid positions (eyes, eyebrows, nose, lips, jaw)

3. Train ResNet-Like Model on Images (in-dim: 125, out-dim: 7)



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(in-dim: 125, out-dim: 7)

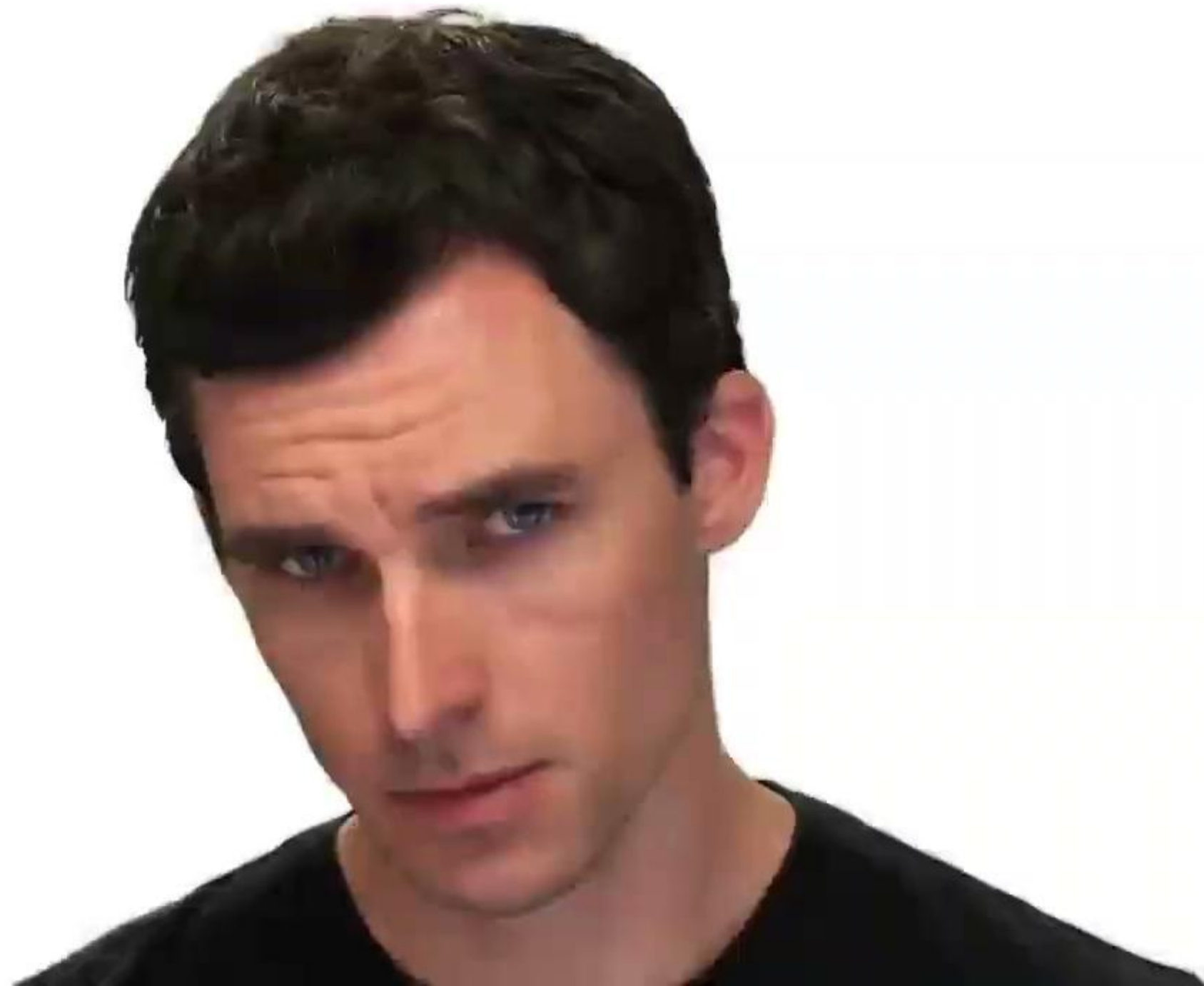


Test Accuracy: 84.55%

Dataset for Temporal FER

RAVDESS: Video frames, classified into 7 emotions

ii.Sad



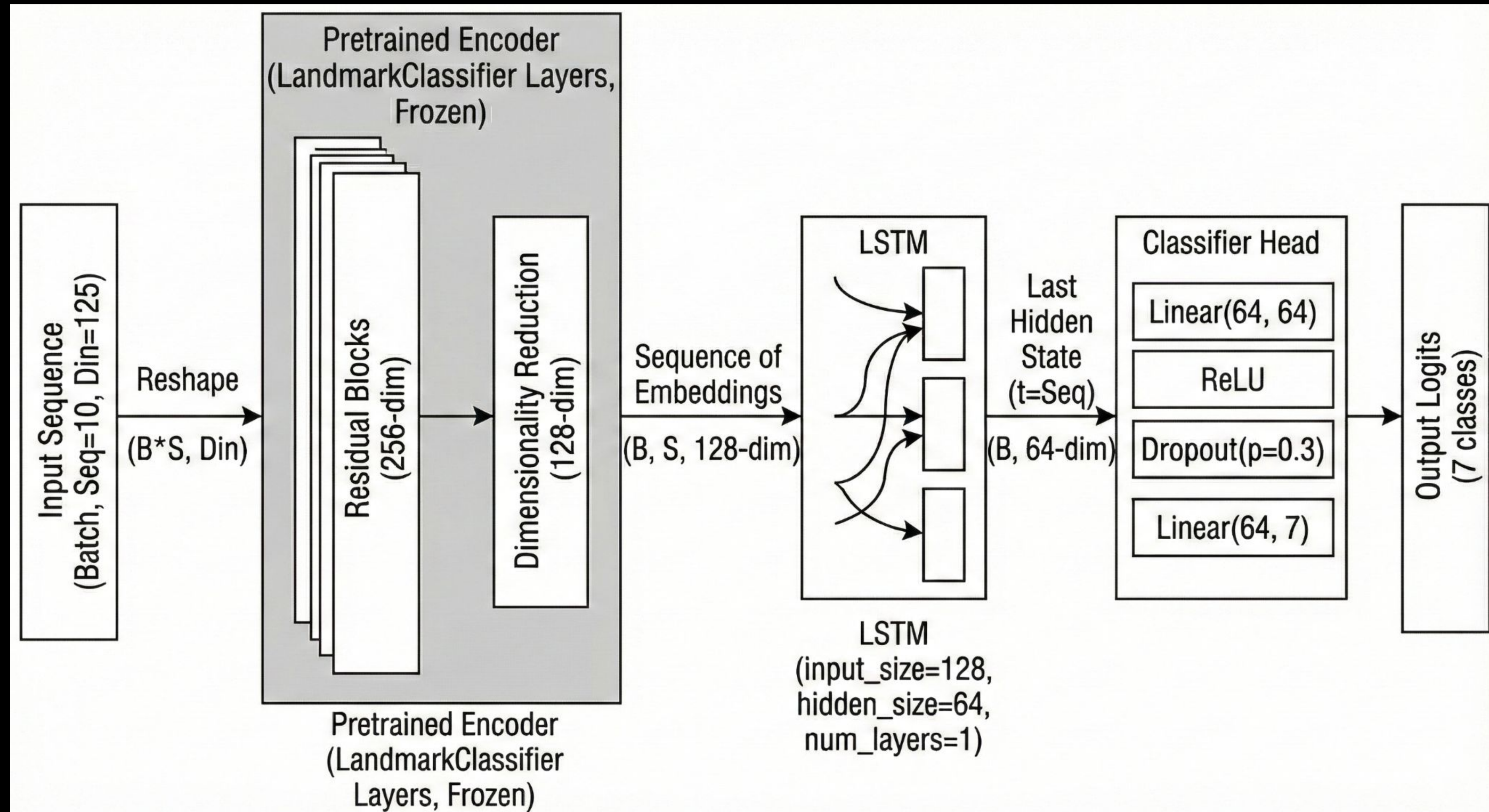
4. Fine-tune Model on Video Frames

(in-dim: [10, 125], out-dim: 7)

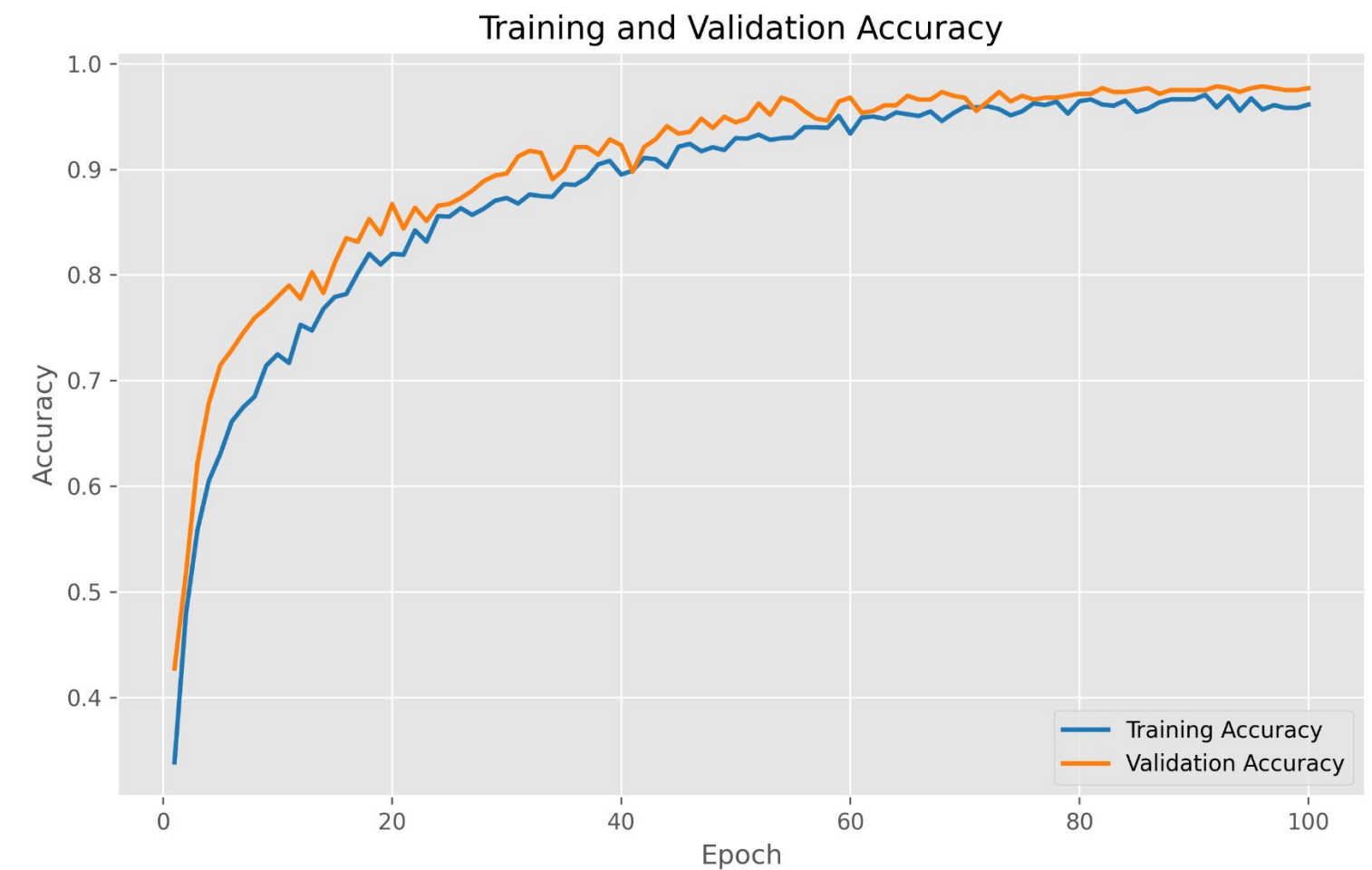
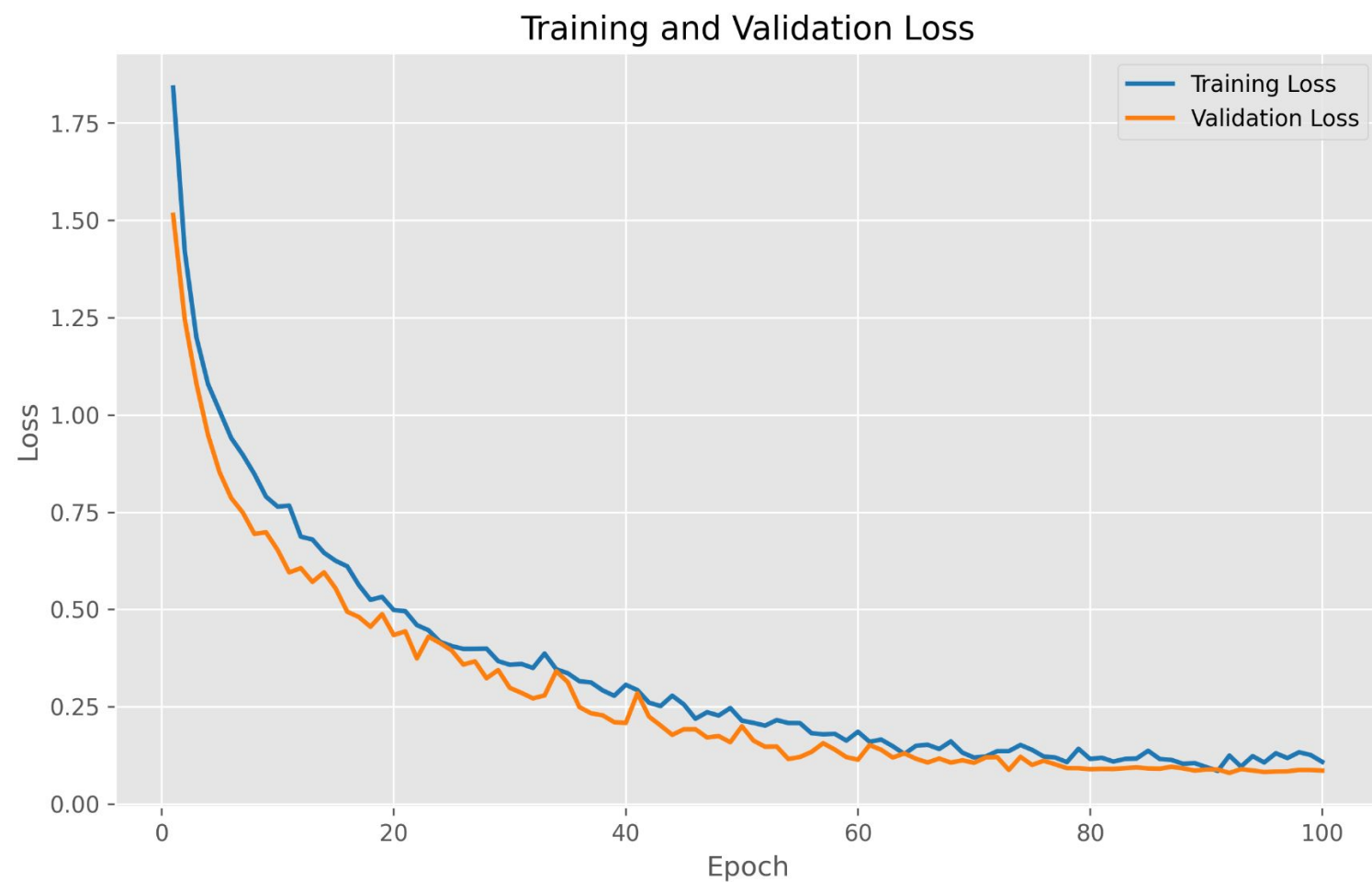
- **Load previously trained ResNet-like model**
- **Remove the final classifier layer**
- **Keep the hidden layers unchanged**
(Keep the layers that can extract facial information from single image)
- **Add new LSTM layers (with 2 FC layers) for sequential input (video dataset)**
- **Fine-tune this model on the video dataset**
(Put a very small learning rate on the previous hidden layers)

4. Fine-tune Model on Video Frames

(in-dim: [10, 125], out-dim: 7)



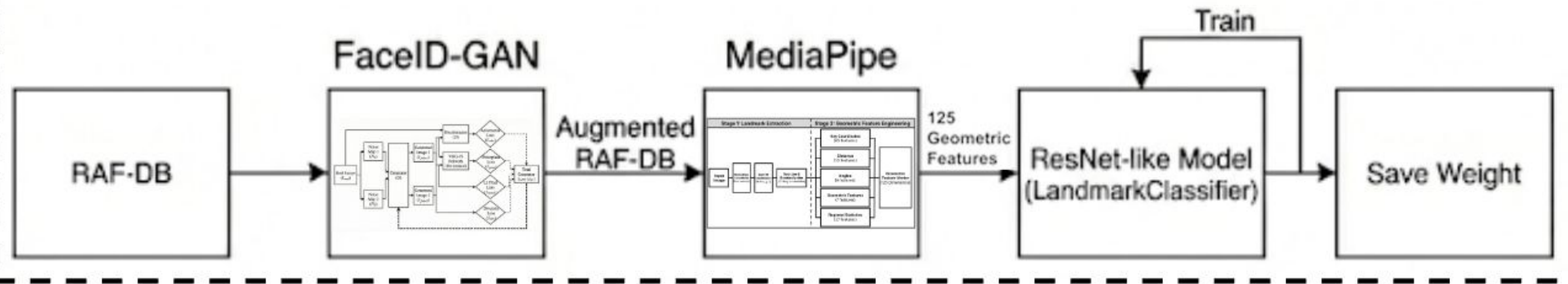
4. Fine-tune Model on Video Frames (in-dim: [10, 125], out-dim: 7)



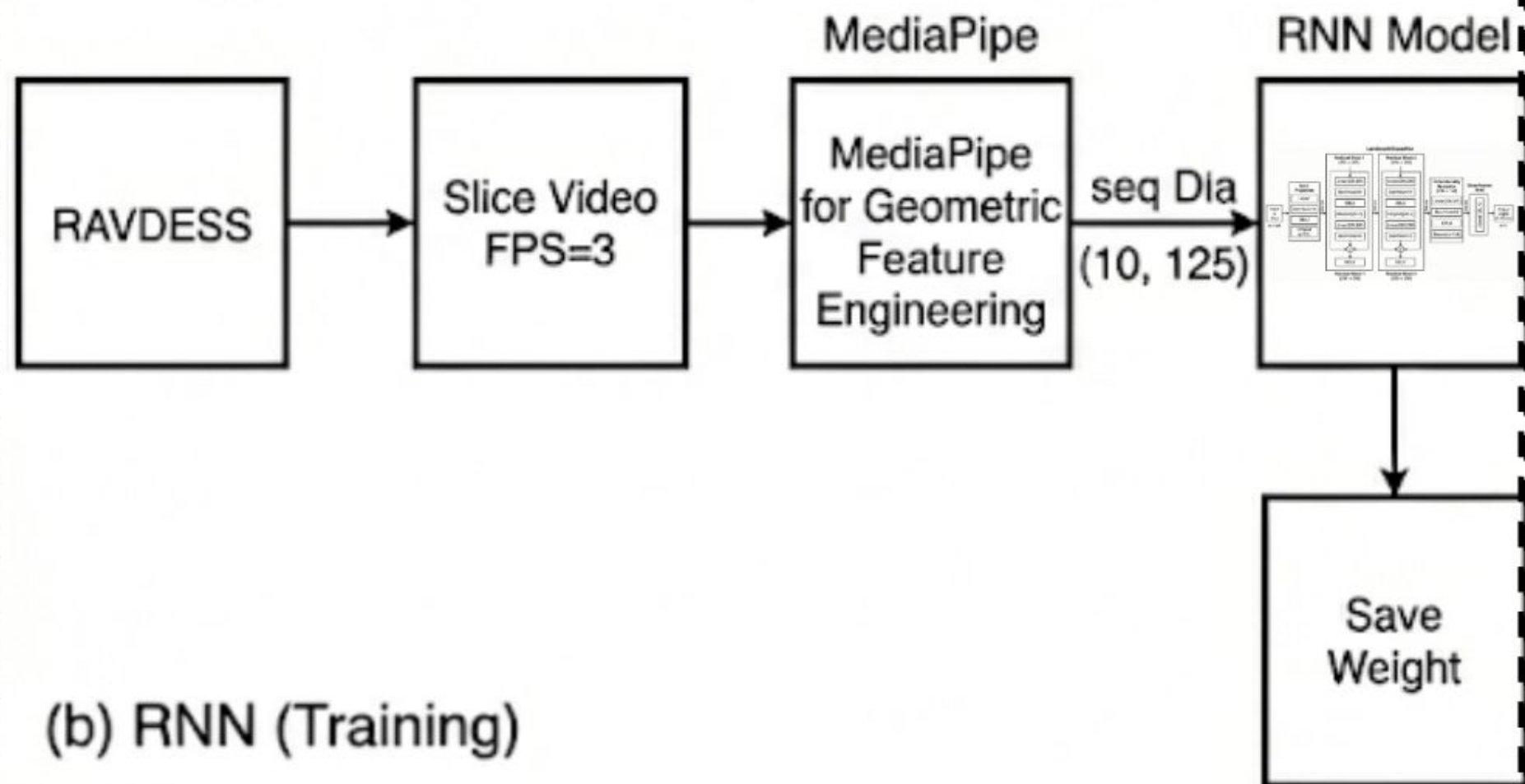
Test Accuracy: 97.85%

Overview of the Training and Inference Pipeline

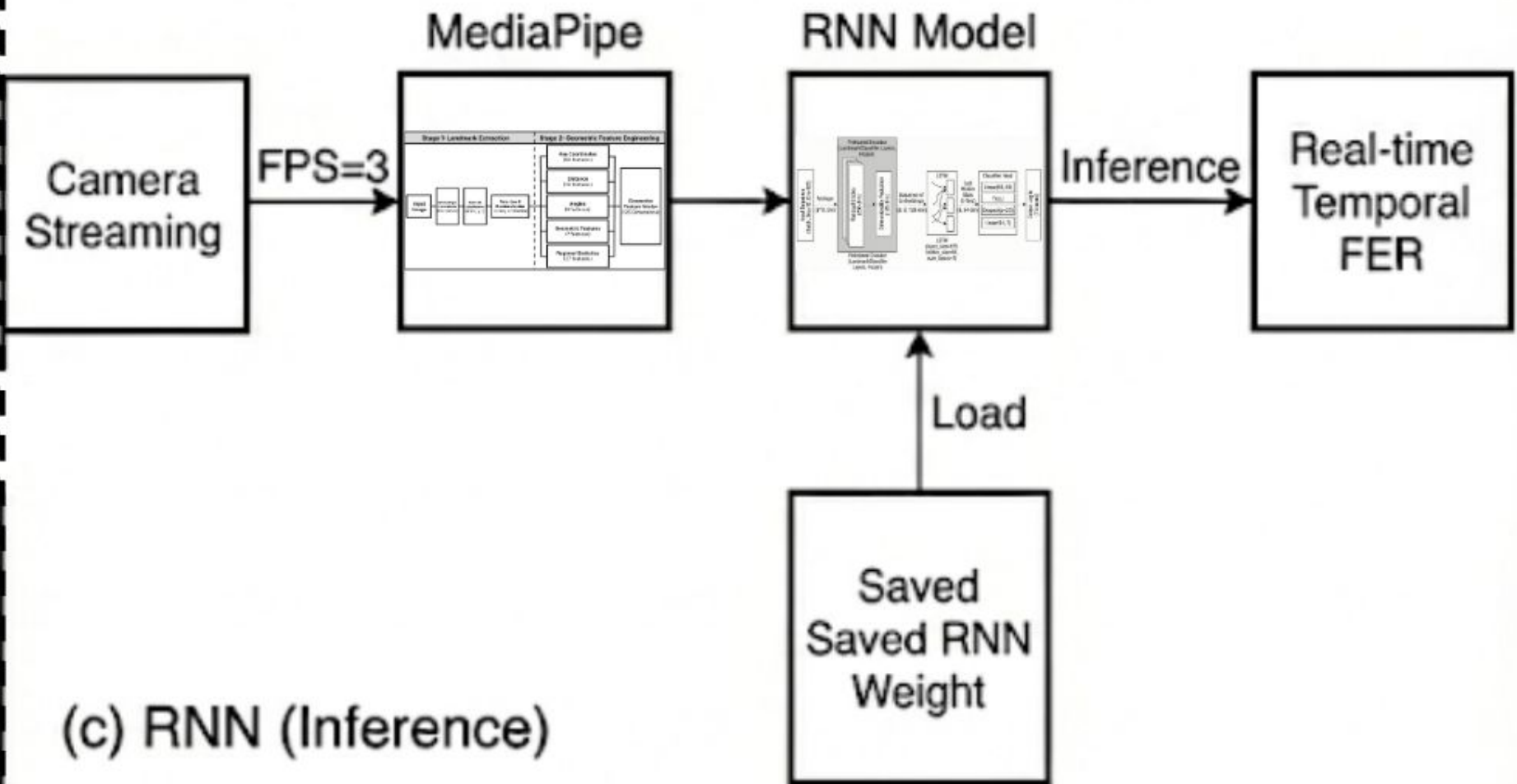
(a) Single Image ResNet-Like (Training)



(b) RNN (Training)



(c) RNN (Inference)





Demo



Thank you